

Groundline v2

An AI brand-intelligence analyst that shows its sources

Reads everything said about Weave and its competitors · alerts on risk · writes cited briefs · a second model fact-checks every claim

A3 collected the data. A4 thinks about it.

BEFORE / AFTER

Assignment 3 collected the data. Assignment 4 thinks about it.

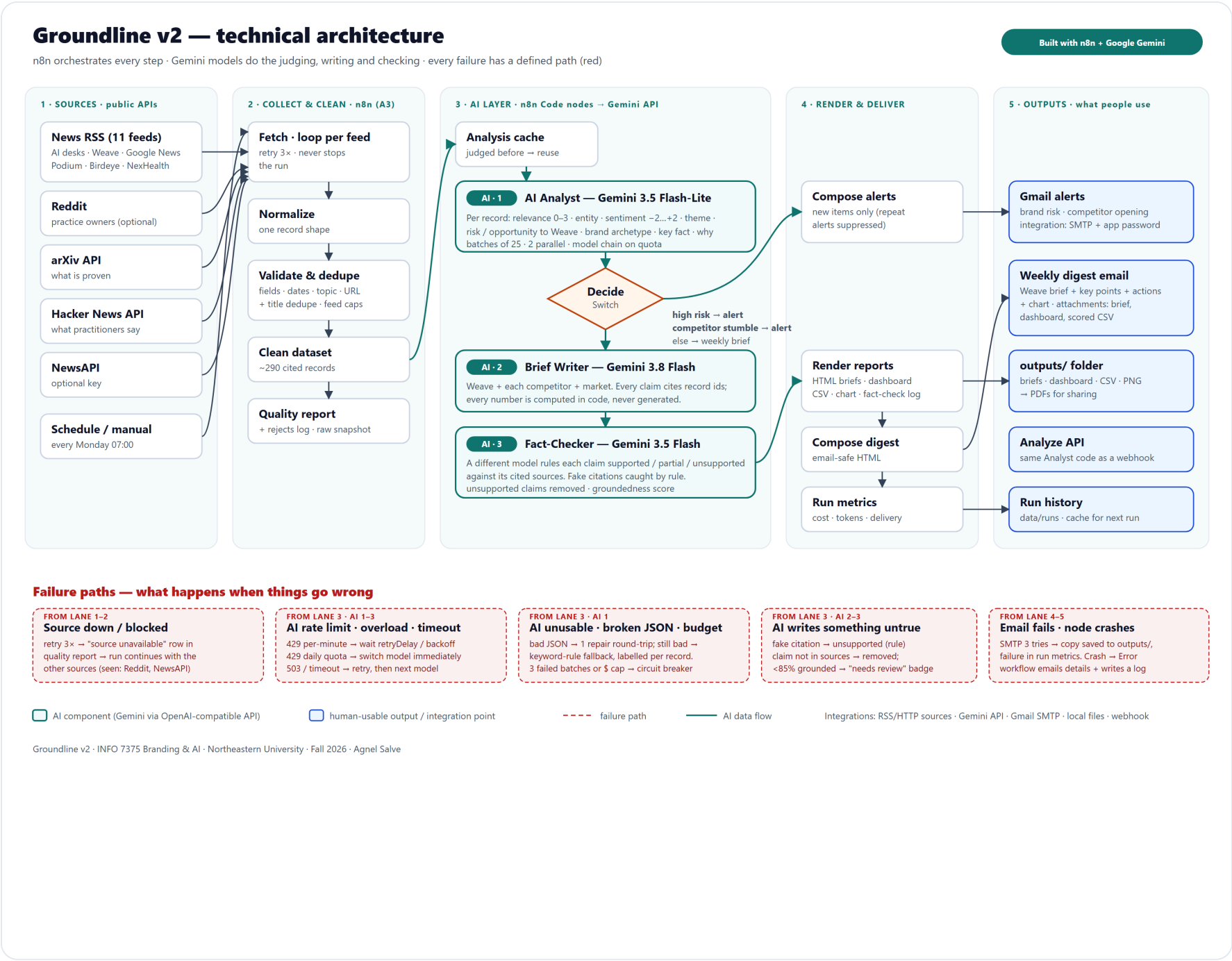
Same pipeline underneath — v2 adds the intelligence layer on top.

	A3 • GROUNDLINE DATA PIPELINE	A4 • GROUNDLINE V2 BRAND-INTELLIGENCE AGENT
What it does	Collects, validates and de-duplicates data	Judges every record, decides, alerts, writes briefs, fact-checks them, delivers
AI	None — keyword rules only	3 Gemini models with separate jobs (judge · write · check), Muse Spark optional
Output	CSV / JSON dataset + quality report	Gmail digest + risk & opportunity alerts, cited HTML/PDF briefs, dashboard, scored CSV, chart
Brand focus	9% of records about Weave; no competitors	Weave + Podium, Birdeye, NexHealth...: share of voice, sentiment, archetype
Relevance	Keyword match — an F1 race 'podium' counts as Podium	AI relevance check removed 13 off-topic items
Trust	Every record cited	Every claim cited, fact-checked by a second model, unsupported claims removed, groundedness score
Error handling	Retry each source 3×	+ backoff, quota-aware model switching, JSON repair, rule fallback, circuit breaker, \$ cap, error workflow
Scale	Not tested	Tested 1 → 2,000 requests; breaking points found and one fixed
Cost tracking	—	Tokens, cost, retries and fallbacks recorded for every run

- A3: 220 cited records, keyword rules, no output a person reads.
- A4: three Gemini models with separate jobs — **judge, write, check**.
- Outputs land in a real Gmail inbox: digest, risk alerts, opportunity alerts.
- Built for Weave's brand / data team (my Madison Intelligence-Agent project).

Say: “Last time I built the pipe. This time the pipe reads, decides and writes — and it has to prove every sentence.”

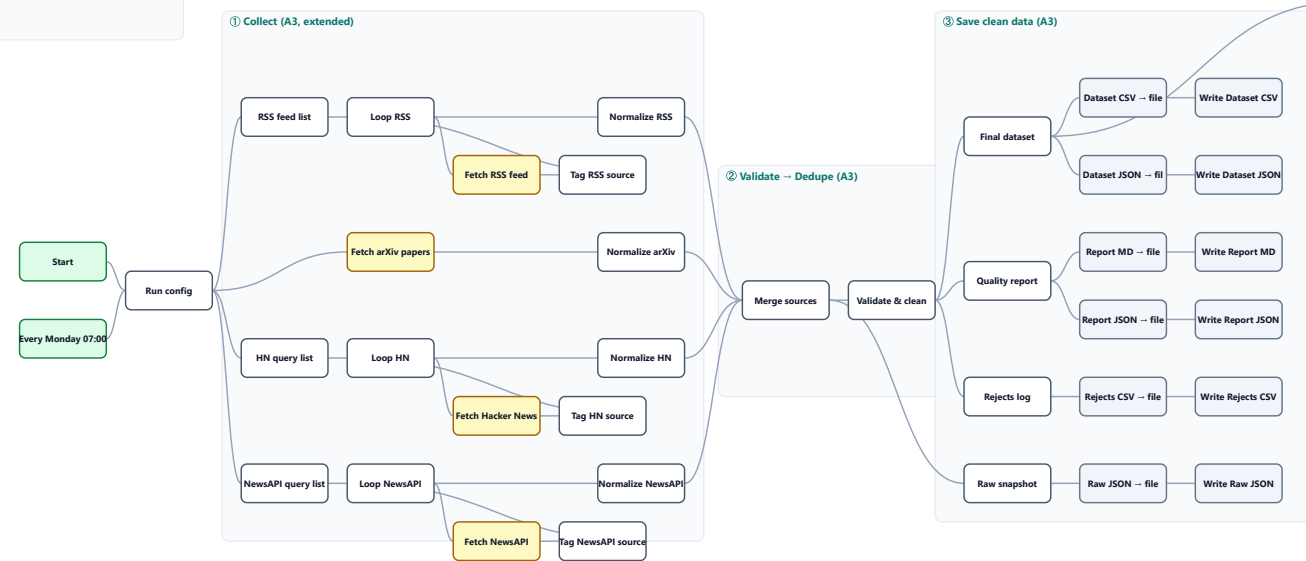
Architecture: five lanes, every failure has a path



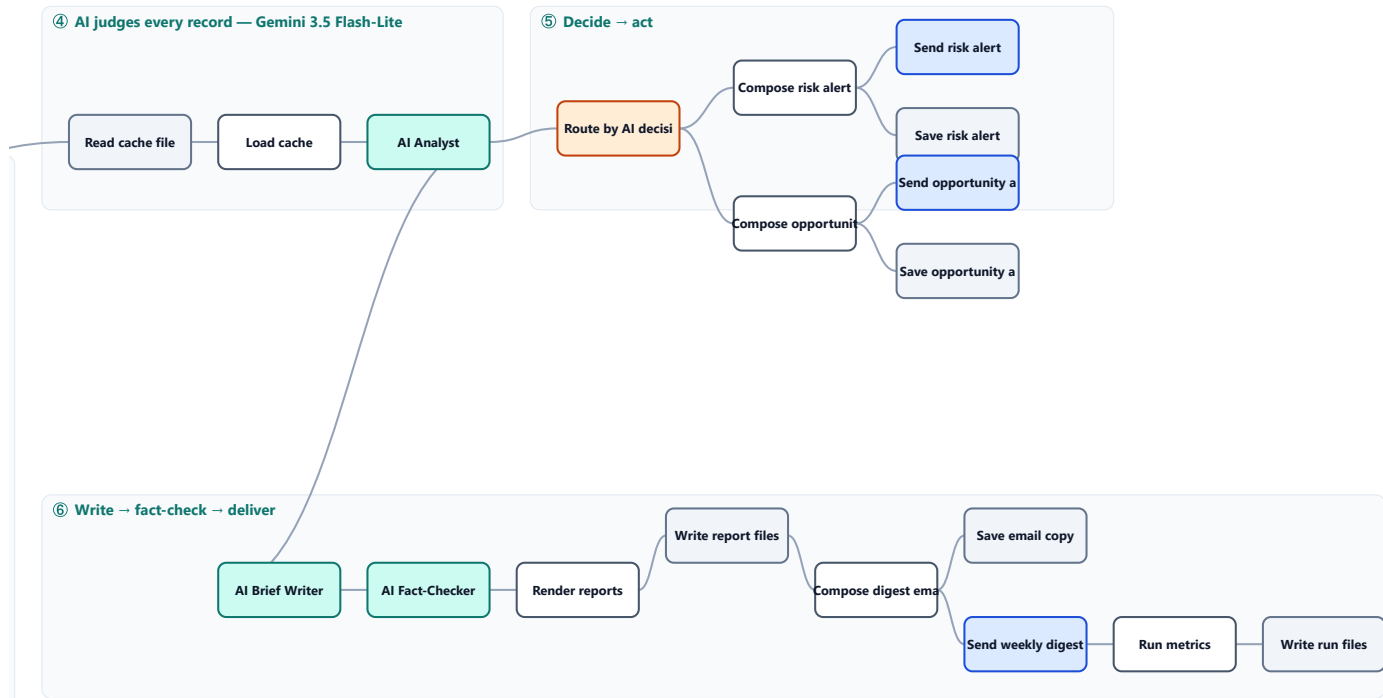
57 n8n nodes, one importable file

A3 · collect, validate, save

Groundline v2 — Brand Intelligence Agent



A4 · judge → decide → alert · write → fact-check → deliver



- Teal: the three AI Code nodes · orange: the Switch decision · blue: Gmail · yellow: source fetches.
- Two more workflows: an **Error handler** and the **Analyze API** webhook used for load tests.
- Code lives in workflow/src + workflow/lib; one build script makes the JSON.

Say: “Everything is in workflow_v2.json — import it and press Execute.” (Canvas drawn from the real workflow file.)

Gemini judges every record — and throws out the noise

OUTPUTS/DATA/2026-10-09_SCORED_RECORDS.CSV • 294 ROWS • OPENS IN EXCEL

Every record with the AI's judgement

ENTITY	RELEVANCE	SENTIMENT	THEME	RISK TO WEAVE	OPPORTUNITY FOR WEAVE	ARCHETYPE	ROUTE	TITLE
Weave	3	0	financials	low	none		digest	Form 4 Weave Communications Inc For: 8 October By Investing.com
Weave	3	0	financials	low	none		digest	A holdings correction at Weave Communications (WEAV) follows Tyler New
Weave	3	-1	legal_regulatory	high	none	Outlaw	urgent_risk	WEAV Stock Alert: Halper Sadeh LLC is Investigating Whether Weave Comm
Weave	3	-1	financials	low	none		digest	Weave Communications Q2 2026 Earnings: Modest EPS Beat Fails to Spark
Weave	3	0	financials	low	none		digest	Weave (NASDAQ: WEAV) holder lines up 32K-share NYSE sale
Weave	3	-1	financials	medium	none		digest	Raymond James Downgrades Weave Communications to Market Perform
Weave	3	-1	financials	medium	none		digest	Raymond James Downgrades Weave Communications to Market Perform From O
Weave	3	-1	financials	medium	none		digest	Raymond James Downgrades Weave Communications(WEAV.US) to Hold Rating,
Weave	3	0	m_and_a	low	none		digest	Raymond James cuts Weave Communications stock rating on buyout deal
Weave	3	2	m_and_a	low	high	Ruler	opportunity	Weave Communications to Be Acquired by Francisco Partners for \$650M
Weave	3	-1	legal_regulatory	medium	none		digest	Weave Communications Investor Alert: Kahn Swick & Foti, LLC Investigat
Weave	3	0	m_and_a	low	none		digest	Weave Communications (WEAV) Stock Fair Value Falls After Analysts Alig
Weave	3	-1	financials	medium	none		digest	Loop Capital Downgrades Weave Communications to Hold From Buy, Adjusts
Weave	3	-1	m_and_a	low	none		digest	Loop Capital downgrades Weave Communications stock on acquisition news

- 294 records: relevance 0–3, entity, sentiment –2...+2, risk, opportunity, archetype, key fact, reason.
- **14 off-topic items removed** — F1 and endurance-racing “podium” stories that A3's keyword rules kept.
- Cache: second run judged 289 of 294 from cache — step time 41 s → 4 s.
- Exported as a spreadsheet anyone can open.

Say: “The AI isn't summarising — it's making 294 small judgements, each with a stated reason.”

High risk → an email the same run

GROUNDLINE • BRAND INTELLIGENCE

Brand risk: 1 new item need attention

Groundline flagged these items as a **risk to Weave's brand**. Each one links to its source so you can verify before acting.

Weave • Business Wire • 2026-09-19 • legal regulatory

WEAV Stock Alert: Halper Sadeh LLC is Investigating Whether Weave Communications, Inc. is Obtaining a Fair Price for its Shareholders

Risk: high Opportunity: none Negative reported

What happened: Halper Sadeh LLC is investigating whether Weave Communications, Inc. is obtaining a fair price for its shareholders.

Why it was flagged: Legal investigations pose severe risks to Weave's stock value and shareholder confidence.

Record GL-6e17f34a • judged by ai:gemin

Decision rule: relevance ≥ 2 and risk_to_weave = high (or a very negative Weave item rated medium).
Run 20261009T045254Z • alerts are sent once per record (repeats are suppressed).

- Switch rule: relevance ≥ 2 and risk_to_weave = high → **risk alert**.
- Real catch: a law firm investigating the \$650M take-private price.
- Competitor stumble or market shift → **opportunity alert** (e.g. ServiceTitan dropping Podium; AI receptionists failing patients).
- Already-alerted items are remembered — no repeat pages.

Say: “This is the decision step: it doesn't just label things, it routes them and acts.”

Every claim cited — and checked by another model

GROUNDLINE BRAND INTELLIGENCE • WEEK OF 2026-10-09

Weave’s \$650M acquisition by Francisco Partners dominates coverage amid ongoing shareholder legal scrutiny

Weave brand brief • 36 items from 21 sources • 2026-07-13 to 2026-10-09

Needs review • 83% grounded

Weave Communications is transitioning to private ownership following a \$650M acquisition agreement. While the market responded positively to the deal, the brand faces legal investigations regarding the sale process and analyst downgrades.

ITEMS TRACKED 36	AVG. SENTIMENT (−2...+2) +0.44	POSITIVE / NEUTRAL / NEGATIVE 15 / 11 / 10	RISKS • OPENINGS FLAGGED 8 • 3
---------------------	-----------------------------------	--	-----------------------------------

What happened

- Weave Communications agreed to be acquired by Francisco Partners for \$650M, a move intended to take the company private.
df38bbbc bded1b74
- The acquisition announcement triggered a surge in stock price and positive market attention. 5c9ff875 a59114d5
- Weave reported record location growth during its Q2 2026 earnings call prior to the acquisition news. b17a0a67

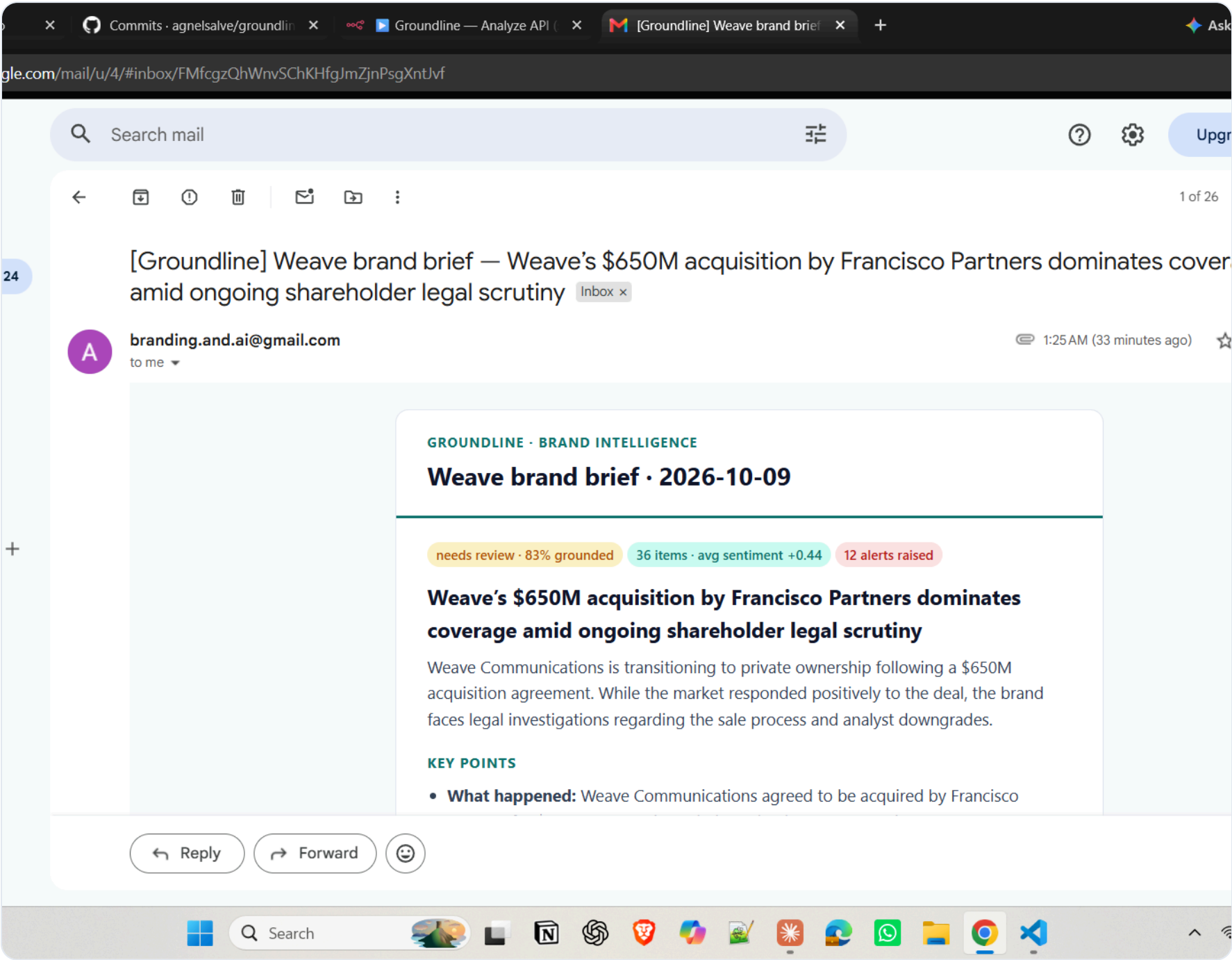
How the brand is perceived

- Market analysts reacted to the acquisition with downgrades, shifting ratings to 'Hold' or 'Market Perform' due to the deal's constraints. 9cef8c1b 40e2d1d8 5779e31b
- The company is increasingly associated with legal and regulatory scrutiny regarding the fairness of the acquisition price. 6e17f34a d1464026

- Brief Writer gets computed STATS + evidence; it must cite record ids for every claim.
- Fact-Checker (different model) rules supported / partial / unsupported.
- Fabricated citations are caught by rule; unsupported claims are **removed** and listed.
- Today: 12 briefs checked, groundedness 54–97%; below 85% = “needs review” badge.

Say: “If the model makes something up, it doesn't reach the inbox — and the brief tells you how much of it was verified.”

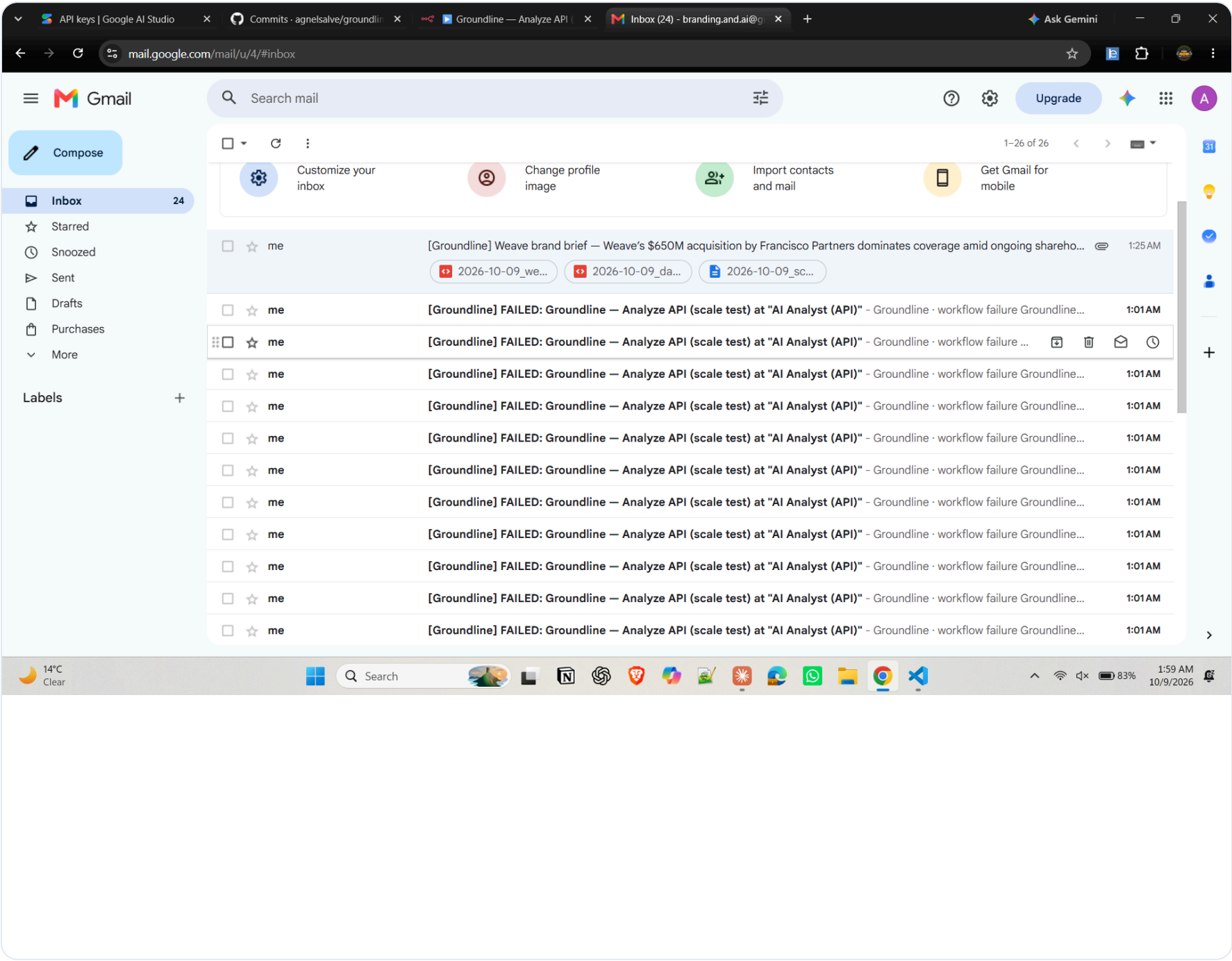
What lands in the inbox



- Weekly digest: headline, key points, 3 actions, chart, competitor one-liners.
- Attached: full brief, dashboard, 294-row spreadsheet.
- 6 real emails delivered today — Gmail message ids in data/runs/.
- Copies of every email are also saved to outputs/.

Say: “A non-technical marketer gets this on Monday morning — no n8n, no JSON.”

When things go wrong, it says so



- Real today: Gemini 3.8 Flash **503 overloaded** and **daily quota** exhausted → automatic switch to the next model.
- Broken JSON → one repair round-trip → keyword-rule fallback, labelled per record.
- Error workflow: a **FAILED email for every crashed run** (inbox row from the scale test) + JSON log; circuit breaker and \$1/run cap.
- No AI key at all → the run still finishes, briefs say “AI offline”.

Say: “I didn't simulate these errors — the free tier produced them, and the log shows each one handled.”

Real load tests: what broke, and the fix

SCALE TESTING • REAL RUNS, 2026-10-09

What happens at 10× and 100× the volume

MODE	REQUESTS	WALL	P50	P95	SUCCESS	RETRIES	HTTP ERRORS	COST	PEAK N8N
AI, before fix	1	14.38s	14.4s	14.4s	0%	3	—	\$0.000	0.4 GB
AI, before fix	10	1.43s	1.4s	1.4s	100%	0	—	\$0.004	0.4 GB
AI, before fix	50	174.9s	61.4s	174.5s	70%	36	500×15	\$0.014	0.9 GB
AI, after fix	50	79.03s	14.5s	78.7s	100%	61	—	\$0.019	1.0 GB
AI, after fix	100	156.6s	34.5s	156.0s	100%	165	—	\$0.038	1.3 GB
AI, batch ×10	5 ×10	7.8s	3.6s	7.8s	100%	0	—	\$0.015	0.4 GB
n8n only	1	0.11s	0.1s	0.1s	100%	0	—	\$0.000	0.4 GB
n8n only	10	0.45s	0.4s	0.4s	100%	0	—	\$0.000	0.4 GB
n8n only	50	1.9s	1.3s	1.9s	100%	0	—	\$0.000	0.7 GB
n8n only	100	3.52s	2.3s	3.4s	100%	0	—	\$0.000	1.4 GB
n8n only	250	10.36s	5.6s	9.9s	100%	0	—	\$0.000	2.9 GB
n8n only	500	17.62s	11.4s	17.0s	100%	0	—	\$0.000	5.5 GB
n8n only	1000	39.34s	24.6s	37.7s	100%	0	—	\$0.000	10.2 GB
n8n only	2000	85.03s	49.7s	83.8s	95.6%	0	500×88	\$0.000	20.6 GB

Breaks first: n8n's Code runner

10 tasks at a time by default; a task waiting >60 s is dropped → 15 × HTTP 500 at 50 concurrent AI requests.
Fix: runner concurrency 50, queue timeout 300 s → 0 errors at 50 and 100.

Then: Gemini free-tier quota

Per-minute limits absorbed by 61–165 retries; daily quota (~20 req/model on Flash) → automatic model switch. Pro models: 0 free quota.

Then: memory

~10 MB per in-flight request; 2,000 burst → 21 GB and 4.4% failures. Platform ceiling here ≈ 1,000 burst / 25 req/s.

Production

Weekly pipeline: 24/7-ready today. High-volume API: paid Gemini tier + n8n queue mode + batching (10× faster per record). ≈ \$9–11/month at 1,000 requests/day.

- 50 concurrent AI requests → 15 × HTTP 500: n8n runs 10 Code tasks at a time.
- Fix: runner concurrency 50, queue timeout 300 s → 0 errors at 50 and 100.
- n8n alone: 1,000-request burst 100%; 2,000 → 4.4% fail, 21 GB.
- ≈ \$0.038 per 100 requests → ≈ \$9–11/month at 1,000/day.

Say: “The first thing to break wasn't the AI — it was n8n's code runner. Finding that is the point of testing.”

Business value and what's next

GROUNDLINE V2 • EXECUTIVE SUMMARY • INFO 7375 ASSIGNMENT 4

An AI brand-intelligence analyst that shows its sources

Every agent answer, traceable to a source. Agnel Salve · Northeastern University · Fall 2026

Built with n8n + Google Gemini

Problem

Weave's brand and data teams track reputation and competitors by hand across news, analyst notes, legal notices and practitioner chatter — slow, and easy to miss the story that matters. AI summaries could do it faster, but nobody can trust them, because nothing ties a claim back to a source.

Solution approach

- **Collect** ~500 items a run from 17 public feeds — Weave, 5 competitors, AI-market news, research, Reddit, Hacker News (A3 pipeline, extended).
- **Judge** every item with Gemini 3.5 Flash-Lite: relevance, sentiment, risk, opportunity, brand archetype, key fact.
- **Decide & act:** high risk → instant Gmail alert; competitor stumble → opportunity alert; repeats suppressed.
- **Write** cited briefs with Gemini 3.8 Flash — numbers computed in code, never generated.
- **Fact-check** every claim with a *different* model; unsupported claims are removed and the brief is scored.
- **Survive failure:** retries, model chain on quota, JSON repair, circuit breaker, budget cap, error workflow.

Business value

- Replaces an estimated **4–6 analyst hours a week** of reading, writing and source-checking for about **\$0.16 of AI per run**.
- Risks reach the inbox the run they appear — e.g. the shareholder-investigation alert on the \$650M take-private.
- Every sentence links to its source and carries a groundedness score, so the team can act on it.
- At 1,000 API requests/day: ≈ **\$9–11 a month** in AI (measured per-request cost).

Current performance (measured 20261009)

294

RECORDS JUDGED PER RUN

301s → 86s

COLD → CACHED RUN

\$0.16

AI COST PER RUN AT LIST PRICE (\$0 ON FREE TIER)

87%

MEDIAN GROUNDEDNESS, 12 BRIEFS FACT-CHECKED (54–97%)

13

NOISE ITEMS THE AI FILTERED OUT

6

REAL EMAILS DELIVERED TO GMAIL (3 OF 3 TYPES)

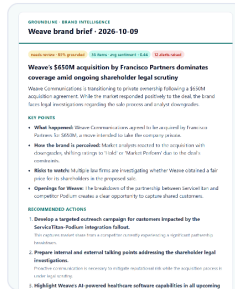
100%

SUCCESS AT 100 CONCURRENT AI REQUESTS

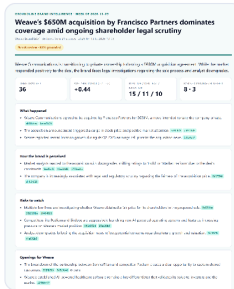
1,000

REQUEST BURST N8N HANDLES (100%)

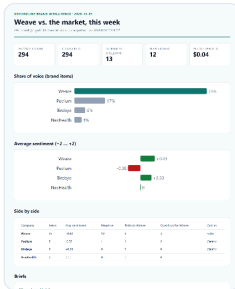
Sample outputs



Weekly digest in Gmail



Cited Weave brief (HTML/PDF)



Weave vs. competitors dashboard

All numbers from real runs on 2026-10-09 — see `scale_test_results.md` and `data/runs/`. Costs at paid Gemini list prices; the free tier billed \$0. "Analyst hours" is an estimate.

- Replaces an estimated 4–6 analyst hours a week for ≈ \$0.16 of AI per run.
- Every sentence traceable to a source → a team can act on it.
- Next: paid Gemini tier + n8n queue mode for the API; route fix (entity ≠ Weave for opportunities).
- Then: feed the cited records into Madison's Intelligence Agent.

Say: "That's Groundline: cheaper than an analyst's afternoon, and it shows its work."

Backup plan

Before presenting

- Run `node scripts/check-setup.mjs` — confirms each Gemini model answers.
- Run once the night before; the cache makes the live run ~1–2 min.
- Keep `outputs/` open in a browser tab: briefs, dashboard, alerts.
- Keep this PDF and the Figma board open.

If something breaks live

- **Gemini quota / 503** — that is the demo: show the model switch in `outputs/run_summary.md`.
- **No internet** — show yesterday's real outputs and the Gmail inbox screenshots.
- **n8n won't start** — open `workflow_v2.json` in the repo and walk this PDF.
- **Email delayed** — open the saved copy in `outputs/emails/`.

Key points to land: (1) three models with separate jobs, (2) every claim cited and independently checked, (3) real decisions → real emails, (4) honest scale limits with a fix, (5) $\approx$ \$0.16 a run.